TABLE VI: Normal Log Types with Representative Examples from Different Software Systems.

Log Type	Representative Example(s)
General Operational Event	*[WAL] [2025-11-28T09:15:03.887Z] INFO web-gateway request=req-482913 method=GET endpoint=/api/profile status=200 backend_calls=3 duration=84ms* *[EESL] Jul 18 09:02:01 EDU-MGMT-SYS INFO task=attendance_sync course=COMP3120 records=186 processed=186 status=COMPLETED* *[SIML] {"timestamp": "2024-06-24T08:05:15.310Z", "level": "INFO", "message": "job=inventory_reconcile store=Store-07 records=12486 updated=12486 status=COMPLETED", "context": {}, "label": "Normal"}*
System Update and Maintenance	*[WAL] [2025-11-28T09:15:10.902Z] INFO maintenance-service node=web-node-06 patch=v3.8.14 install=SUCCESS restart=SUCCESS service_state=RUNNING* *[CDL] 081109 102412 21 INFO db.schemaManager: SCHEMA_UPDATE: node=db-node-03 schema_from=4.12 schema_to=4.13 tables=28 rollback=0 status=COMPLETED* *[FALL] [04.05 15:05:24] facline.exe - controller=PLC-A17 config=PLC-CFG-2025.04 deployment=SUCCESS validation=PASS diagnostics=PASS*
User Action	*[WAL] [2025-11-28T09:15:01.234Z] INFO auth-service user=U-19384 action=LOGIN src_ip=172.18.6.24 session=SID-82471 result=SUCCESS* *[ULML] [2025-01-22T08:14:15] INFO equipment-booking user=student-2187 action=BOOK equipment=LAB-MIC-07 booking=EXP-6118 start=14:00 end=16:00 status=CONFIRMED* *[EPL] Oct 3 12:10:45 shop1-230-105-77 kernel[0]: ORDER: customer=C-582104 action=SUBMIT_ORDER order=ORD-20250318-9126 items=4 amount=286.40 currency=USD status=ACCEPTED*
Network and Communication	*[CDL] 081109 105623 34 INFO db.connectionManager: CONNECTION_ESTABLISHED: channel=db-node-02→db-node-05 protocol=replication state=CONNECTED heartbeat_latency=11ms* *[MSL] 17/05/06 08:33:02 INFO api-gateway: trace=tr-7f21d9 request=req-93127 src=order-service dst=inventory-service status=200 duration=37ms* *[IOTL] {"log": "2024-02-23T08:30:12.432Z INFO device-comm node=SN-042 gateway=GW-03 protocol=MQTT session=ESTABLISHED rssi=-61dBm keepalive=60s", "label": "Normal"}*
Handled Error / Warning	*[WAL] [2025-11-28T09:15:03.887Z] WARN cache-service cache_lookup key=user-profile:5821 result=TIMEOUT fallback=DATABASE fallback_status=SUCCESS total_latency=132ms* *[MSL] 17/05/06 08:36:12 WARN payment-service: request=req-48217 src=payment-service dst=authentication-service result=SOFT_TIMEOUT retry=1 retry_status=SUCCESS latency=146ms* *[IOTL] {"log": "2024-02-23T08:35:12.432Z WARN device-comm node=SN-113 sequence=8821 packet=MISS retransmission=1 retransmit_status=SUCCESS packet_loss=0", "label": "Normal"}* *[SIML] {"timestamp": "2024-06-24T08:07:58.420Z", "level": "WARN", "message": "sku=SKU-81942 terminal=T-04 conflict=VERSION_MISMATCH resolution=LATEST_VERSION sync_status=SUCCESS", "context": {}, "label": "Normal"}*
Process and Service Lifecycle	*[MSL] 17/05/06 08:32:17 INFO container-service: container=inventory-service-7f94d8 node=k8s-worker-03 event=START readiness=PASS startup=2.8s* *[WAL] [2025-11-28T09:15:05.041Z] INFO auth-service service=auth-service event=INITIALIZED pid=18342 profile=production-v5 workers=12 state=RUNNING* *[FALL] [04.05 15:07:31] facline.exe - process=WC-17 station=WS-07 state=RUNNING temperature_controller=READY displacement_sensor=READY interlock=READY*
Database Operation	*[CDL] 081109 105623 34 INFO db.queryExecutor: QUERY_EXECUTED: query=Q-82931 shard=04 operation=SELECT rows=18426 latency=47ms status=OK* *[FTL] [2023-12-10-10.24.13.345678] INFO transaction-service txn=TXN-98273145 operation=COMMIT verification=PASS replicas_ack=2 state=COMMITTED* *[EPL] Oct 3 12:03:50 shop1-230-105-77 kernel[0]: DB: order=ORD-573184 operation=INSERT inventory_reservation=COMMITTED payment_reference=COMMITTED txn_state=COMMITTED*
System Health and Resource Monitoring	*[CDL] 081109 110959 50 INFO db.monitor: HEALTH_CHECK: node=db-node-07 cpu=58.0% buffer_pool=64.0% replication_lag=8ms cluster_state=HEALTHY* *[FALL] [04.05 15:10:01] facline.exe - controller=A12 motor_temperature=48.6°C vibration=1.8mm/s cycle_time=7.42s state=NORMAL* *[IOTL] {"log": "2024-02-23T08:35:12.432Z INFO telemetry gateway=GW-08 nodes_online=126 nodes_total=126 delivery_ratio=99.7% avg_rssi=-58dBm state=ONLINE", "label": "Normal"}*
File Operation	*[WAL] [2025-11-28T09:15:09.120Z] INFO file-service file=/opt/web/config/runtime.yml action=READ pid=4821 bytes=18432 checksum=VALID status=SUCCESS* *[FTL] [2023-12-10-10.28.21.567890] INFO file-service file=txn-audit-20250318.csv action=WRITE records=184291 destination=/archive/2025/03 encryption=AES256 status=SUCCESS* *[FALL] [04.05 15:12:18] facline.exe - file=weld-line-A-20250318.log action=TRANSFER src=PLC-A07 dst=monitor-01 bytes=481920 checksum=PASS*
Audit Event	*[FTL] [2023-12-10-10.20.01.124567] AUDIT audit-service operator=OP-381 action=VIEW transaction=TXN-6728134 role=FraudReview modified=0 timestamp=15:42:18* *[ULML] [2025-01-22T08:19:15] AUDIT lab-admin approver=LM-07 action=GRANT_ACCESS equipment=HV-03 user=researcher-214 authorization=APPROVED* *[SIML] {"timestamp": "2024-06-24T08:12:49.214Z", "level": "AUDIT", "message": "supervisor=U-104 action=STOCK_ADJUST sku=SKU-66213 old=38 new=42 reason=CYCLE_COUNT status=APPROVED", "context": {}, "label": "Normal"}*
Security Monitoring	*[WAL] [2025-11-28T09:15:01.234Z] SECURITY auth-service gateway=auth-gw-02 failed_logins=0 unauthorized_requests=0 token_anomalies=0 verdict=CLEAN* *[FTL] [2023-12-10-10.26.17.456789] SECURITY fraud-service txn=TXN-98273145 device_fingerprint=PASS identity=PASS payment_token=PASS decision=ALLOW* *[MSL] 17/05/06 08:35:46 SECURITY payment-service: src=billing-service dst=payment-service mtls=SUCCESS tls=1.3 certificate=71A93F state=ESTABLISHED*
User Input Validation	*[WAL] [2025-11-28T09:15:07.512Z] INFO web-gateway request=req-55218 validation=PASS username_length=12 email_format=VALID password_policy=PASS missing_fields=0* *[FTL] [2023-12-10-10.22.10.234567] INFO transaction-service txn=TXN-721983 validation=PASS currency=HKD amount=1280.00 account_format=VALID token=VALID* *[SIML] {"timestamp": "2024-06-24T08:15:23.345Z", "level": "INFO", "message": "purchase_order=PO-20250318-084 supplier=SUP-042 items=16 invalid_items=0 warehouse=WH-03 validation=PASS", "context": {}, "label": "Normal"}*

TABLE VII: Anomalous Log Types with Representative Examples from Different Software Systems.

Log Type	Representative Example(s)
Runtime Exception	*[WAL] [2025-11-28T09:15:38.266Z] ERROR UserProfileController NullPointerException request=req-58214 session=sid-8217 endpoint=/api/profile* *[EESL] Jul 18 10:56:12 EDU-MGMT-SYS ERROR scheduler job=JOB-2187 exception=IllegalStateException semester=2025S1 course=COMP3120 slot=TUE-14:00* *[EPL] Oct 3 12:50:33 shop1-230-105-77 kernel[0]: ORDER: order-service SerializationException order=ORD-728194 event=order-confirmation user=C-58102*
Authentication and Access Violation	*[FTL] [2023-12-10-10.48.44.789012] WARN access-service access denied operator=OP-174 resource=settlement-records role=FraudReview required_role=FinanceAdmin txn=TXN-719842* *[WAL] [2025-11-28T09:15:16.441Z] WARN auth-service authentication failed user=U-48192 src=185.47.22.61 attempts=5 window=18s account_status=LOCKED* *[ULML] [2025-01-22T08:41:45] WARN access denied user=student-4287 equipment=HV-03 authorization=NONE laboratory=LAB-05*
Service Failure	*[MSL] 17/05/08 10:19:59 ERROR payment-service: service=payment-service pod=payment-7f94d8 readiness_probe=FAILED consecutive_failures=6 state=UNAVAILABLE* *[WAL] [2025-11-28T09:16:18.010Z] ERROR auth-service service=auth-service instance=auth-03 startup=FAILED health_status=DOWN response_code=503* *[EPL] Oct 3 12:50:33 shop1-230-105-77 kernel[0]: API: service=order-sync state=UNRESPONSIVE order=ORD-618293 pending_events=284 last_heartbeat=19:04:32*
Timeout and Communication Failure	*[MSL] 17/05/07 09:06:12 ERROR api-gateway: request timeout src=order-service dst=inventory-service request=req-93184 elapsed=3000ms retries=3 trace=tr-8fa271* *[IOTL] {"log": "2024-02-23T08:55:12.432Z device-comm heartbeat timeout node=SN-084 missed=6 duration=94s rssi=-87dBm"* *[CDL] 081109 192710 12 INFO db.connectionManager: CONNECTION_TIMEOUT: replication timeout primary=db-node-02 replica=db-node-05 transaction=TX-91827 ack=MISSING elapsed=5000ms* *[FTL] [2023-12-10-11.00.56.456789] ERROR api-service gateway timeout txn=TXN-817234 provider=bank-gw-02 elapsed=60s status=PENDING*
Performance and Resource Degradation	*[WAL] [2025-11-28T09:15:49.209Z] WARN web-gateway host=web-node-06 endpoint=/api/v2/checkout issue=UPSTREAM_TIMEOUT p95_latency=2.7s p99_latency=4.9s timeout_count=63 http_5xx=57 active_sessions=1842* *[CDL] 081109 164322 17 INFO db.performance: RESOURCE_DEGRADATION: host=db-node-07 memory=96.2% swap=81.4% buffer_pool=93.1% active_connections=947* *[SIML] {"timestamp": "2024-06-24T08:22:55.772Z", "level": "ERROR", "message": "terminal=T-07 disk_used=99.2% free_space=0.8% sync_cache=BLOCKED store=Store-12", "context": {}, "label": "Abnormal"}*
Data Integrity Violation	*[FTL] [2023-12-10-10.36.32.123456] ERROR transaction-service amount mismatch txn=TXN-98273145 debit=8420.00 settlement=8240.00 currency=HKD account=AC-58193* *[CDL] 081109 170518 35 INFO db.storage: CHECKSUM_MISMATCH: block=blk-51892 primary=db-node-03 replica=db-node-08 checksum_primary=8f21a4 checksum_replica=91c82e* *[SIML] {"timestamp": "2024-06-24T08:30:18.092Z", "level": "WARN", "message": "stock mismatch sku=SKU-72184 terminal_count=38 central_count=71 terminal=T-04 store=Store-07", "context": {}, "label": "Abnormal"}*
Malware and Suspicious Execution	*[WAL] [2025-11-28T09:16:09.973Z] ALERT security-service unsigned executable path=/tmp/update-helper user=websvc pid=19482 sha256=7d93a81f... verdict=SUSPICIOUS* *[EPL] Oct 3 12:57:10 shop1-230-105-77 kernel[0]: SECURITY: malware detected file=invoice-update.exe signature=Trojan.Generic uploader=U-28417 action=QUARANTINED* *[FTL] [2023-12-10-11.14.18.567890] ALERT security-service process=powershell.exe parent=txn-worker.exe user=svc-payment dst=198.51.100.27:443 policy=BLOCKED*
Concurrency and Synchronization Failure	*[CDL] 081109 170518 35 INFO db.queryExecutor: DEADLOCK_DETECTED: transaction1=TX-18472 transaction2=TX-18491 table=customer_account victim=TX-18491 node=db-node-04* *[MSL] 17/05/07 09:00:02 WARN inventory-service: concurrent update conflict reservation=RSV-7281 pod1=inventory-17 pod2=inventory-21 version1=84 version2=85* *[FALL] [04.05 15:50:27] facline.exe - synchronization mismatch plc=PLC-A07 robot=RC-03 line=A station=07 cycle=CY-18427 plc_state=READY robot_state=BUSY*
Dependency Failure	*[MSL] 17/05/08 10:13:27 ERROR checkout-service: service=checkout-service dependency=pricing-service state=UNAVAILABLE attempts=3 endpoint=pricing-service:8080* *[WAL] [2025-11-28T09:16:18.010Z] ERROR auth-service shared library load failed service=auth-service host=web-node-03 library=libssl.so.3 path=/usr/lib64* *[EPL] Oct 3 12:50:33 shop1-230-105-77 kernel[0]: API: external API service=logistics provider=ship-gw protocol_version=1.4 expected=2.0 order=ORD-581943 status=REJECTED*
System Crash	*[WAL] [2025-11-28T09:16:28.990Z] FATAL system-service host=web-node-08 process=java pid=18422 signal=SIGSEGV core_dump=/var/crash/core.18422* *[CDL] 081109 203504 27 INFO db.nodeManager: NODE_CRASHED: host=db-node-04 event=KERNEL_PANIC uptime=17d04h device=nvme0n1 reboot=INITIATED* *[FALL] [04.05 15:45:15] facline.exe - controller=PLC-B12 line=B station=12 state=EMERGENCY_STOP error_code=HW-COMM-07 cycle=CY-72184*
Numerical and Threshold Deviation	*[IOTL] {"log": "2024-02-23T08:40:12.432Z WARN telemetry sensor=SN-119 gateway=GW-06 pressure=8.74bar lower_limit=4.50bar upper_limit=6.00bar timestamp=14:28:17", "label": "Abnormal"}* *[FALL] [04.05 15:40:12] facline.exe - station=WS-07 batch=WB-1842 weld_temperature=428.6°C upper_limit=410.0°C cycle=WC-5184 terminal_count=92* *[WAL] [2025-11-28T09:15:49.209Z] WARN api-rate-limit endpoint=/api/order p95_latency=2860ms threshold=1500ms window=5m requests=2841 node=web-node-04*
Scheduling and Orchestration Conflict	*[ULML] [2025-01-22T10:01:45] ERROR resource conflict equipment=MIC-07 booking1=EXP-4812 booking2=EXP-4861 overlap=14:00–15:30 laboratory=LAB-03* *[MSL] 17/05/07 09:13:05 WARN deployment-service: deployment conflict service=catalog-service deploy_job=deploy-8721 migration_job=mig-441 migration_state=PREEMPTED* *[FALL] [04.05 15:55:11] facline.exe - schedule conflict station=WS-03 job1=JOB-A128 job2=JOB-A131 overlap=92s production_line=A*
Sensor and Industrial Process Anomaly	*[FALL] [04.05 15:40:12] facline.exe - sensor=DS-17 station=WS-04 displacement=14.82mm lower_limit=11.50mm upper_limit=12.50mm cycle=WC-5184* *[FALL] [04.05 15:55:11] facline.exe - station=A09 sequence_violation current=STEP-06 expected=STEP-05 job=JOB-7284* *[IOTL] {"log": "2024-02-23T09:20:12.432Z WARN device-comm gateway=GW-11 node1=SN-144 node2=SN-145 timestamp_offset=18.6s synchronization=FAILED batch=BT-4182", "label": "Abnormal"}*